

CSCN8020 - Reinforcement Learning Programming

Assignment 2: Q-Learning on the Taxi-v3 Environment

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Course: CSCN8020 - Reinforcement Learning Programming

Summary

This project implements Q-Learning, a model-free reinforcement learning algorithm, on the Taxi-v3 Gymnasium environment (500 states, 6 actions). A tabular Q-table is trained for 5,000 episodes using the Temporal Difference update with epsilon-greedy action selection. The code is organised into clear classes for the environment, agent, trainer, experiments, logging, and plotting. Hyperparameter experiments vary the learning rate (alpha) and the exploration rate (epsilon). The best combination (alpha=0.2, epsilon=0.1, gamma=0.9) converges fastest and reaches an average greedy evaluation reward of +7.45. Training progress is logged and visualised with smoothed comparison charts.

Repository

Clone URL: https://github.com/muthuacumen/CSCN8020_Assignment2.git

How to Run

```
git clone https://github.com/muthuacumen/CSCN8020_Assignment2.git
cd CSCN8020_Assignment2
python -m venv .venv && .venv\Scripts\activate
pip install -r requirements.txt
jupyter lab CSCN8020_Assignment2.ipynb    # then: Run All Cells
```

Full run: ~60-90 s. Cell outputs are pre-saved; no re-run needed to view results.